

Customer Churn & Credit Risk Analytics

Data Monetisation Unit - Predictive Analytics Initiative

KEY PERFORMANCE INDICATORS

Total Customers Analysed	Churn Rate	Default Rate	High-Risk Customers
10,000	67.1%	70.1%	6,709

BUSINESS PROBLEM

The bank faces dual challenges: retaining increasingly mobile customers (churn) while managing credit exposure (default). Undetected churn costs an estimated 5-7x more than retention; unmanaged defaults directly impair the loan portfolio. This project delivers early-warning scores for both risks using customer transaction history, product usage, and support interaction sentiment - enabling the Data Monetisation team to prioritise outreach and credit reviews efficiently.

METHODOLOGY

- ETL Pipeline: 10,000 synthetic customer records merged with 5,000 support chat logs. Missing values imputed via median/mode. Tableau/Power BI-ready CSV exported.
- Feature Engineering: Sentiment score (TF-IDF + Logistic Regression), complaint flag, engagement score (recency x frequency x value), high-utilisation flag.
- Models: Logistic Regression (baseline), Random Forest, and XGBoost trained on 80/20 stratified split for both churn and credit risk targets.

MODEL RESULTS - CHURN PROPENSITY

Model	ROC-AUC	Precision	Recall	F1
Logistic Regression	1.000	0.999	0.997	0.998
Random Forest	1.000	1.000	1.000	1.000
XGBoost [BEST]	1.000	1.000	1.000	1.000

MODEL RESULTS - CREDIT RISK (DEFAULT)

Model	ROC-AUC	Precision	Recall	F1
Logistic Regression	0.985	0.978	0.968	0.973
Random Forest	1.000	1.000	1.000	1.000
XGBoost [BEST]	1.000	1.000	1.000	1.000

BUSINESS RECOMMENDATIONS

- Targeted Retention Campaigns**
Deploy XGBoost churn scores in CRM to trigger automated outreach (e.g., loyalty offers) for customers with churn probability > 0.6.
- Early Credit Review Alerts**
Flag customers with default probability > 0.6 for monthly credit review before the 90-day delinquency threshold.
- Sentiment-Driven Support Triage**
Integrate the sentiment model into the contact centre - route negative-sentiment customers to senior agents to reduce escalations.
- Product Personalisation**
Loan-only customers show the highest churn rate; offer cross-sell savings accounts to improve product stickiness and lifetime value.

JD ALIGNMENT - SKILLS DEMONSTRATED

BANK GROUP | EXECUTIVE SUMMARY

Unstructured Data (NLP) -> Chat log sentiment analysis using TF-IDF + LR | Model Building -> Churn propensity & credit scoring with LR / RF / XGBoost | Data Visualisation -> Streamlit dashboard + Tableau/Power BI CSV export | ETL & Data Flows -> Full pipeline: generate -> merge -> clean -> parquet | Providing Insights -> This executive summary with actionable recommendations